Advanced Programming Practice · 21CSC203P

Unit 4 · Session 6 · SLO 1 — Solved Worksheet

Name: AKARSH MURALI AKARSH MURALI | **Reg. No:** RA2511030010067

Unit – 4 : Session – 6 : SLO – 1 | Parallel Computing Paradigms and Key Models

I. Fill in the Blanks

1. Parallel — parallel computing allows multiple processes to run simultaneously.
2. Shared — shared-memory architecture gives all processors one global memory space.

II. Multiple Choice

1. b) Shared Memory — tightly coupled processors communicate through a common address space with low latency, so the shared-memory model suits them best.

III. Key Notes

• Shared memory: tightly coupled, communication via variables in common memory (e.g. multicore CPUs, OpenMP).
• Distributed memory: loosely coupled, each node owns private memory, communication via message passing (e.g. clusters, MPI).